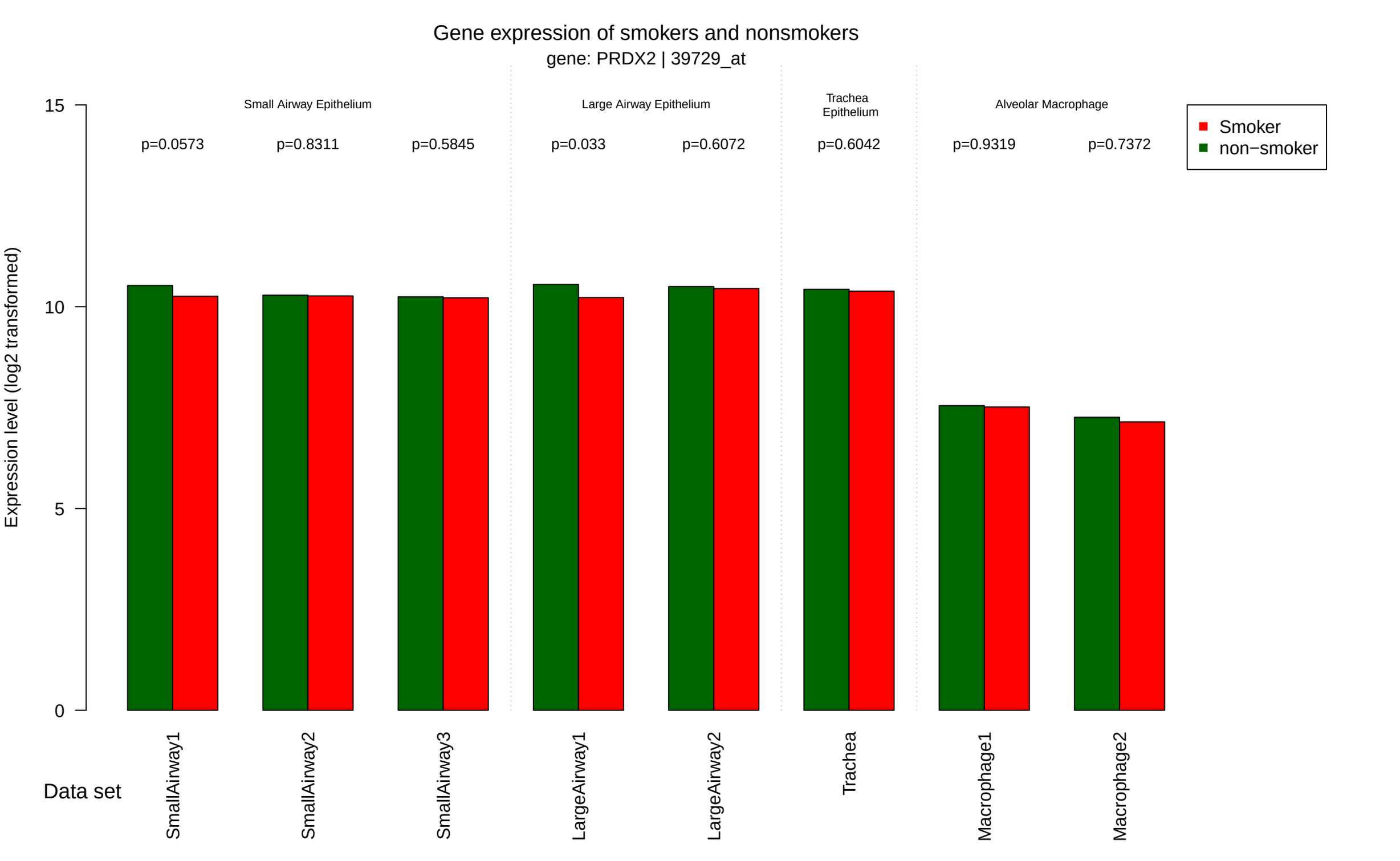
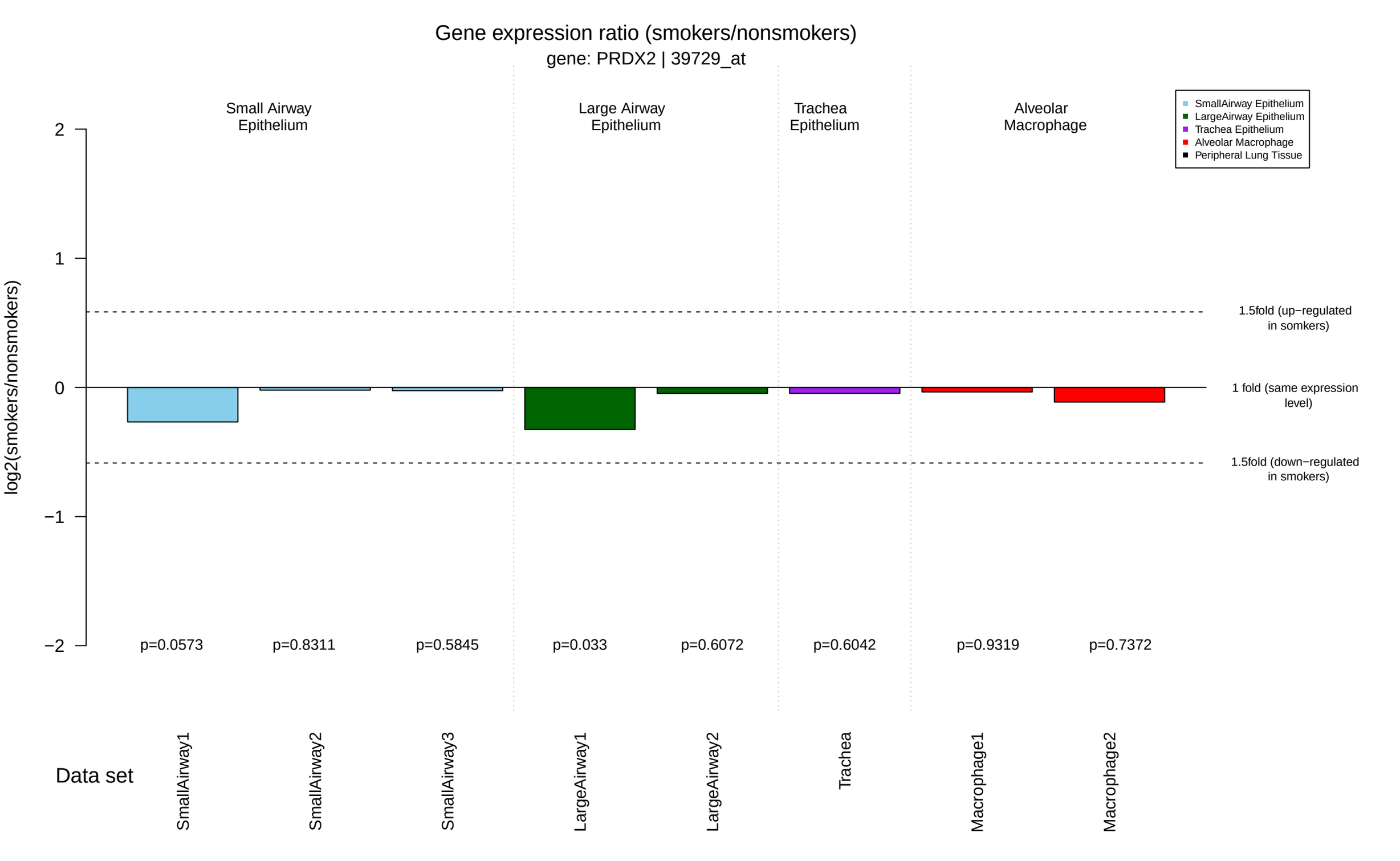
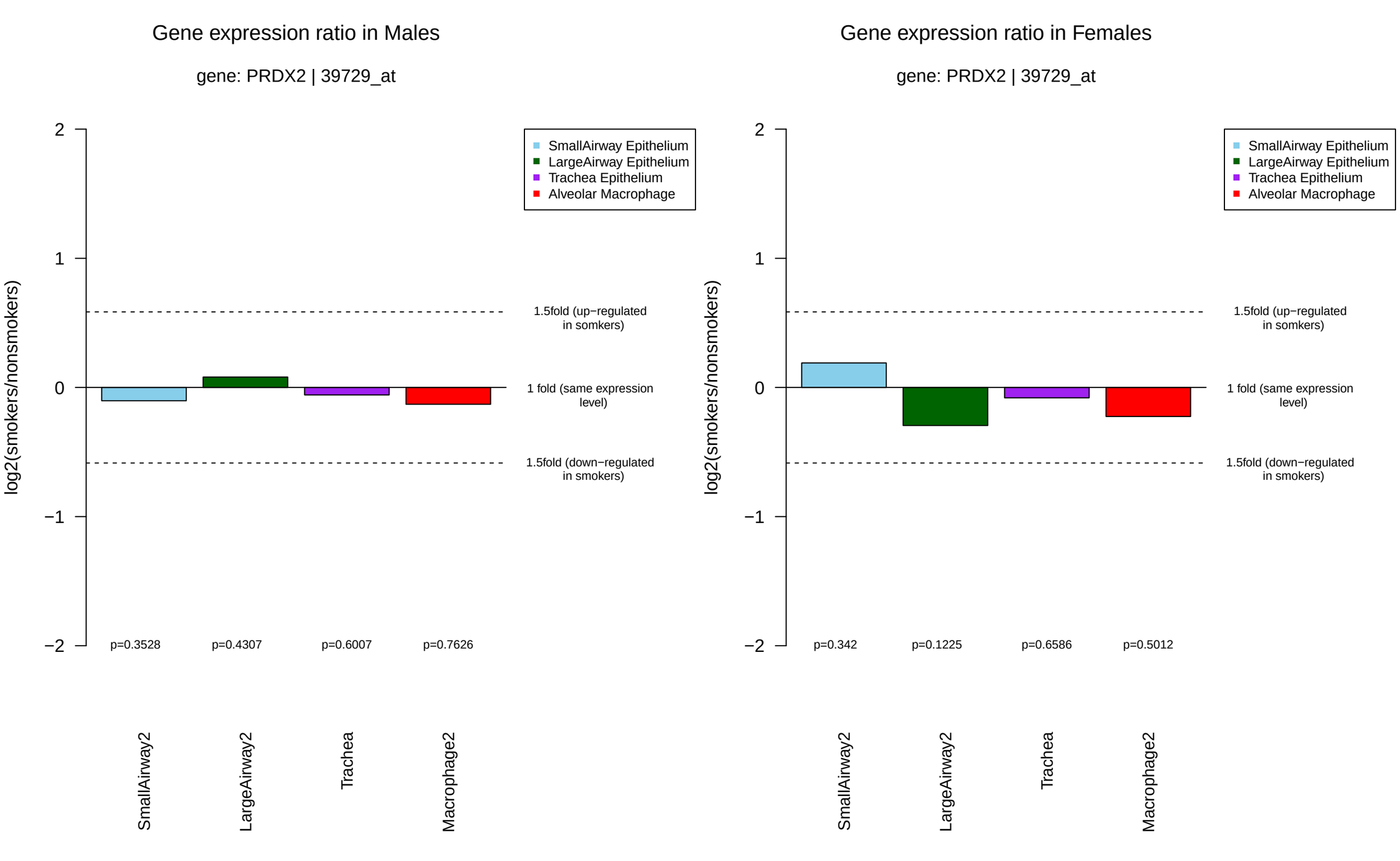
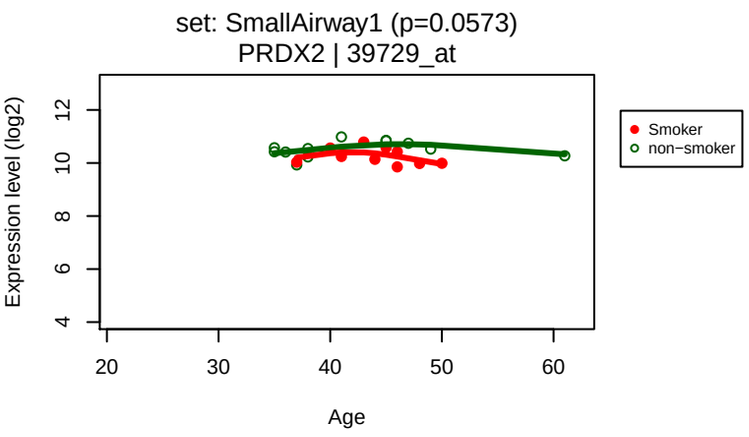
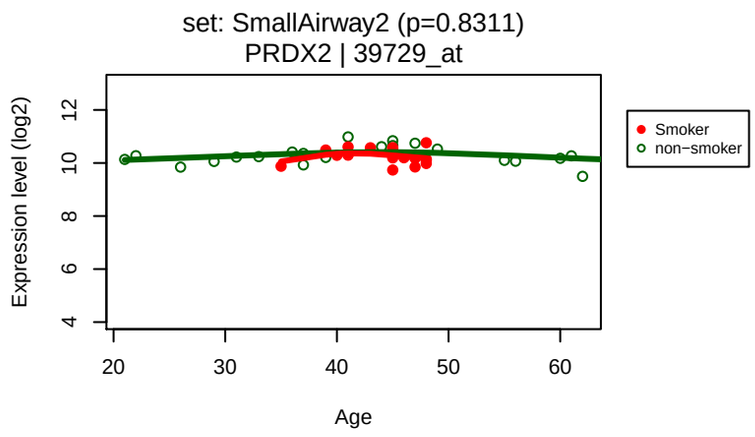
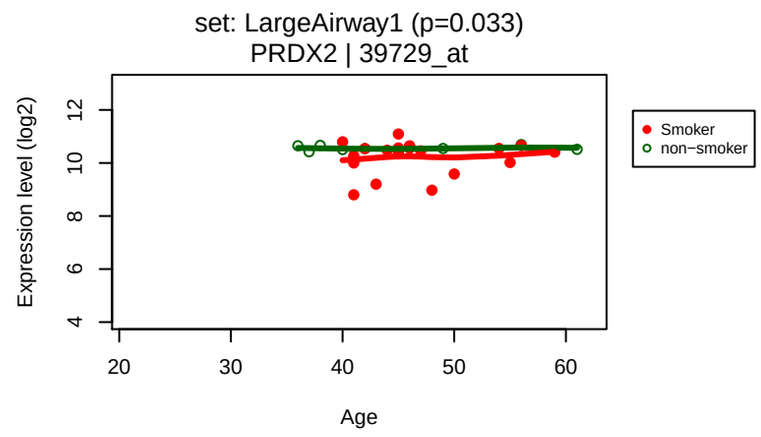

Gene expression of smokers and nonsmokers
gene: PRDX2 | 39729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0573
p=0.8311
p=0.5845
p=0.033
p=0.6072
p=0.6042
p=0.9319
p=0.7372
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

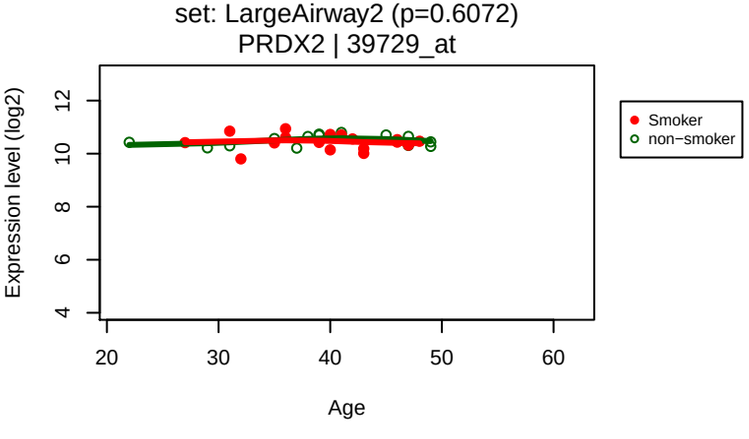

Gene expression ratio (smokers/nonsmokers)
gene: PRDX2 | 39729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0573
p=0.8311
p=0.5845
p=0.033
p=0.6072
p=0.6042
p=0.9319
p=0.7372
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

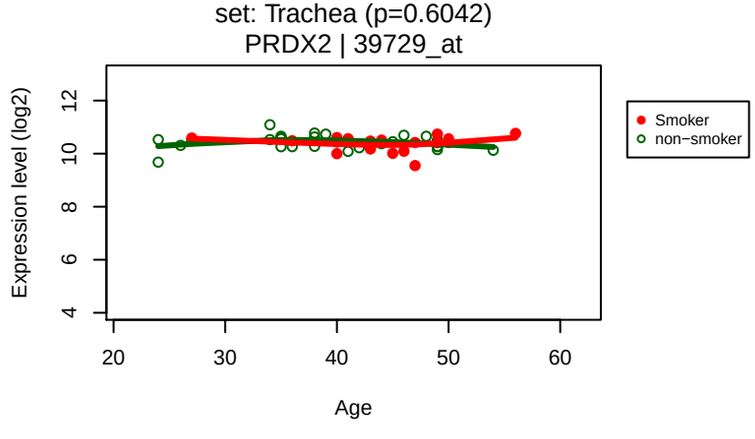

Gene expression ratio in Males
gene: PRDX2 | 39729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3528
p=0.4307
p=0.6007
p=0.7626
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRDX2 | 39729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.342
p=0.1225
p=0.6586
p=0.5012
SmallAirway2
LargeAirway2
Trachea
Macrophage2

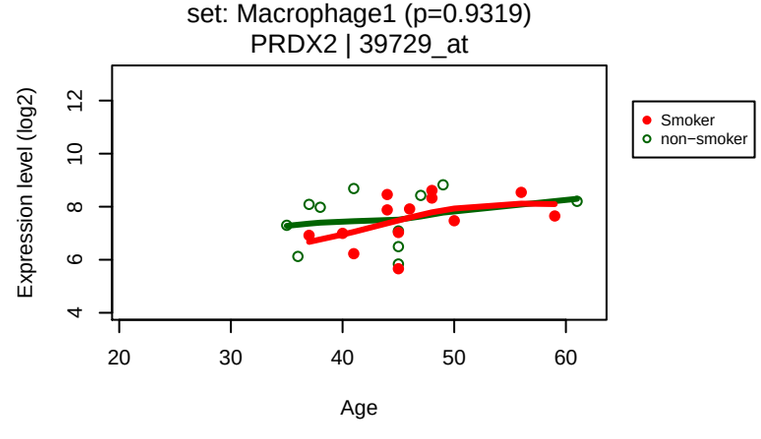

set: SmallAirway1 (p=0.0573)
PRDX2 | 39729_at
Expression level (log2)
Age
Smoker
non−smoker

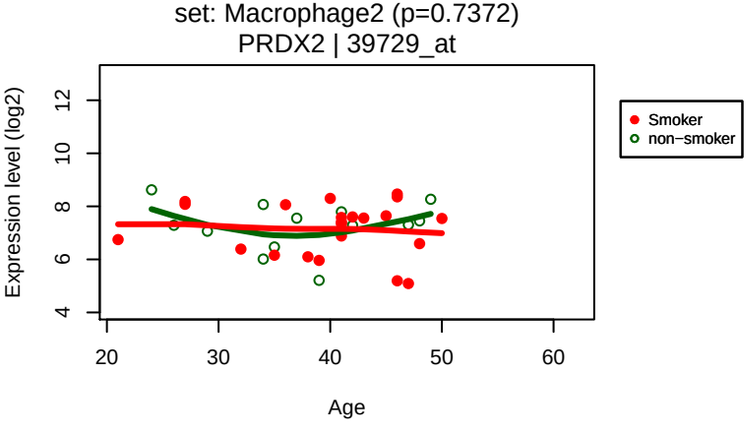

set: SmallAirway2 (p=0.8311)
PRDX2 | 39729_at
Expression level (log2)
Age
Smoker
non−smoker

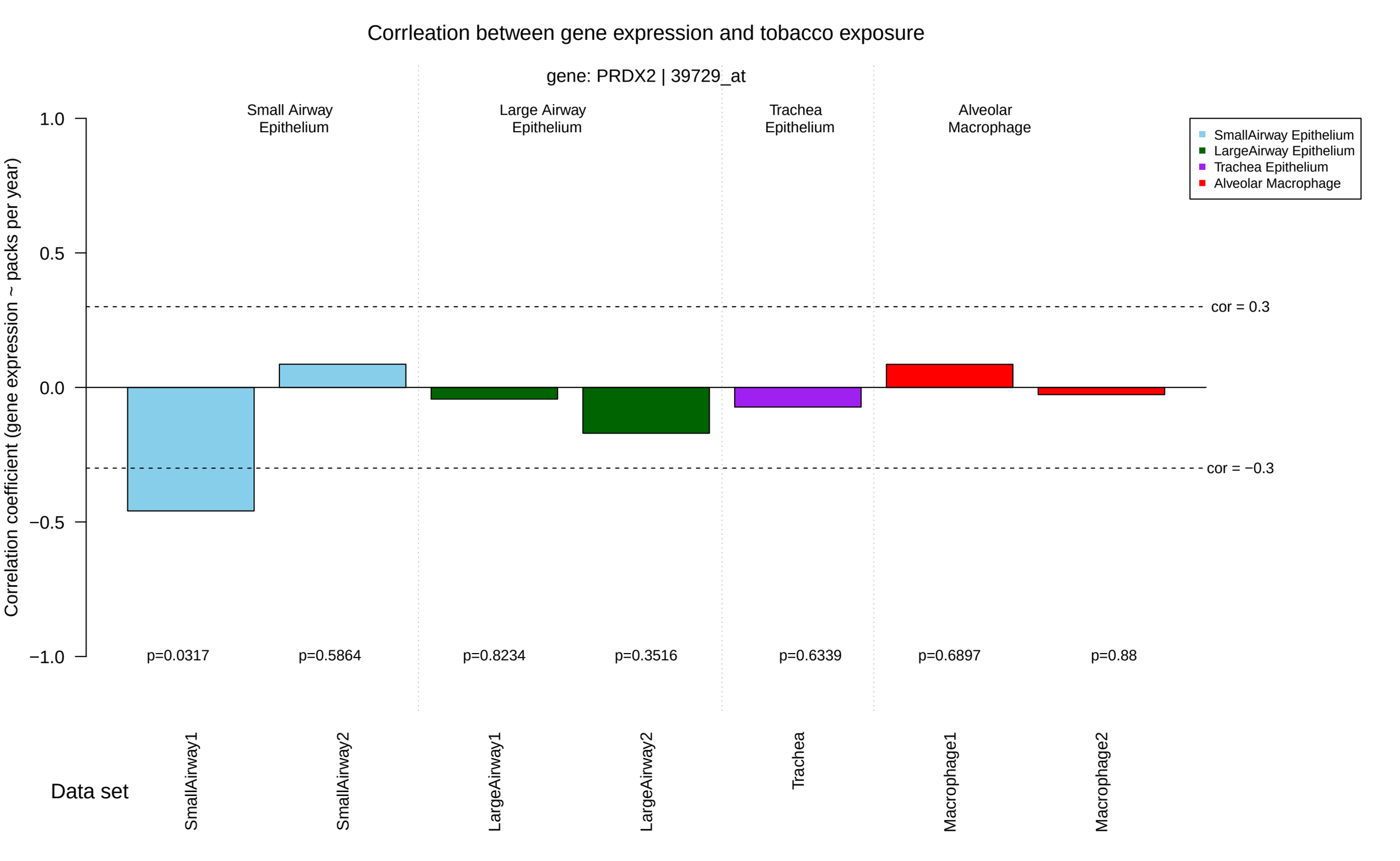

set: LargeAirway1 (p=0.033)
PRDX2 | 39729_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6072)
PRDX2 | 39729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6042)
PRDX2 | 39729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9319)
PRDX2 | 39729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7372)
PRDX2 | 39729_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRDX2 | 39729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0317
p=0.5864
p=0.8234
p=0.3516
p=0.6339
p=0.6897
p=0.88
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2